

Answer all questions.

1. (a) A student wishes to determine the material from which a metal ball bearing is made. He obtains the following values and uses them to determine the density of the metal in the ball bearing.

$$\text{Volume of ball bearing} = 5.6 \pm 0.2 \text{ cm}^3$$

$$\text{Mass of ball bearing} = 45.4 \pm 0.5 \text{ g}$$

- (i) Calculate the density of the ball bearing (in g cm^{-3}) and show that its **percentage** uncertainty is approximately 5%. [3]

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- (ii) Determine the **absolute** uncertainty in the density. [1]

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- (b) The table gives the density of some common metals and alloys.

Metal	Density / g cm^{-3}
Tin	7.3
Stainless steel	7.5
Iron	7.9
Brass	8.2
Nichrome	8.3
Copper	8.9



- (i) Write down possible materials from which the ball bearing is made and explain why it is not possible to determine the exact material. [2]

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- (ii) Identifying the correct material from which the ball bearing is made depends on reducing the uncertainty in density. Explain which of the two values (volume or mass) contributes more to this uncertainty. [2]

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